

Topic 2 Review [100 marks]

1. [Maximum mark: 4] 25M.1.SL.TZ2.4

Construct a logic diagram for the following expression:

$$X = \text{NOT } A \text{ OR } B \text{ AND NOT } C \quad [4]$$

2. [Maximum mark: 2] 25M.1.SL.TZ2.8

A binary number is held in the following 12-bit register:

1	1	1	1	0	0	1	1	1	0	1	0
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- (a) State the hexadecimal representation of this binary number. [1]
- (b) State how many different binary numbers can be represented in a 12-bit register. [1]
3. [Maximum mark: 4] 25M.1.HL.TZ2.2
- (a) Outline the purpose of secondary memory. [2]
- (b) Describe a situation when secondary memory would be used as an extension of primary memory. [2]
4. [Maximum mark: 4] 25M.1.HL.TZ2.3
- Outline **two** possible hardware upgrades to improve a desktop computer's performance. [4]

5. [Maximum mark: 4] 25M.1.HL.TZ3.5
Construct a truth table for the following expression:

$W = C \text{ OR NOT } B \text{ AND NOT } A \text{ OR } B$ [4]

6. [Maximum mark: 2] 24N.1.SL.TZ0.8
Identify **two** advantages of low power consumption digital devices for users. [2]

7. [Maximum mark: 2] 24N.1.SL.TZ0.1
State **two** features of spreadsheets. [2]

8. [Maximum mark: 15] 24N.1.SL.TZ0.11
Laptops are used daily for data storage, web browsing, gaming, and email.

- (a) State **one** precaution a user can take to secure their data in case their laptop is stolen. [1]
- (b) Outline **one** feature of the operating system that is required when running a program on a laptop. [2]
- (c) Justify the decision of the laptop manufacturer to include both wired **and** wireless network connection capability. [4]

A data packet is a basic unit of communication over a computer network.

- (d) Describe the structure of a data packet. [2]
- (e) Outline **three** reasons why protocols are necessary on a computer network. [6]

9. [Maximum mark: 15]

24N.1.SL.TZ0.12

(a) Consider the following expression:

$$(X > 6) \text{ OR } (Y > 3) \text{ AND } ((X + Y) < 20)$$

(a.i) State **all** the Boolean operators in this expression. [1]

(a.ii) State **all** the constants in this expression. [1]

(a.iii) Determine the value of this expression when X is 6 and Y is 6.
Show all your working. [2]

A student plans to create an alarm system for their room. The room has one door and one window.

Three sensors will be used in the system:

- A sensor to detect movement in the room.
- A sensor to detect if the door is locked or unlocked.
- A sensor to detect if the window is open or closed.

The student knows that this practical problem can be expressed in terms of Boolean logic and presented in a truth table.

The student considers the following three inputs:

1. MOTION, where true (1), represents that movement is detected in the room and false (0) represents no movement is detected.
2. DOOR, where true (1), represents that the door is locked and false (0) represents an unlocked door.
3. WINDOW, where true (1), represents that the window is open and false (0) represents a closed window.

A siren will make a warning noise when the door is locked and **either** of the following occur:

- Movement is detected.
- The window is open.

- (b) Copy and complete the truth table for this alarm system.

MOTION	DOOR	WINDOW	SIREN
0	0	0	

[4]

The central processing unit (CPU) executes a program that is stored as a sequence of machine language instructions in primary memory. It does this by repeatedly fetching an instruction from primary memory, decoding that instruction, executing and storing the result.

- (c.i) Identify the CPU register that holds data to be transferred to primary memory. [1]
- (c.ii) Identify the part of the CPU that performs decoding. [1]
- (c.iii) State where the calculations will be executed. [1]
- (c.iv) Explain the role of buses in the execution of a machine language instruction. [4]

10. [Maximum mark: 3] 24M.1.SL.TZ1.9

- (a) Outline the purpose of the memory data register (MDR). [2]
- (b) Outline the role of the arithmetic logic unit (ALU). [1]

11. [Maximum mark: 2] 24M.1.SL.TZ1.10

- (a) State the binary equivalent of the denary number 37. [1]
- (b) State the hexadecimal equivalent of the binary number 10011110. [1]

12. [Maximum mark: 2]24M.1.SL.TZ1.11

Distinguish between **two** types of primary memory. [2]

13. [Maximum mark: 15]24M.1.SL.TZ1.14

(a) Define the NOR Boolean operator. [1]

A car has features that monitor its speed, direction and distance from the car in front.
This is shown in **Figure 1**.

Figure 1: Rules to control car motion

Input	Binary representation	Description
A	0	Car is less than 20 metres from the vehicle in front.
	1	Car is 20 metres or more from the vehicle in front.
B	0	Car is travelling in reverse or stationary.
	1	Car is travelling forward.
C	0	Car speed is more than 130 kilometres per hour.
	1	Car speed is 0–130 kilometres per hour.

For example, if the car is travelling forward, input B would have a binary representation of 1.

(b) Construct a logic diagram with inputs A, B, and C and output Z to represent the following scenario:

Output Z equals 1 when:

- the car is travelling forward AND it is less than 20 metres from the vehicle in front.
- OR

- the car speed is more than 130 km per hour.

[4]

In all other conditions, output Z equals 0.

An additional row (input D) is to be added to assist when the car is in reverse or stationary. Input D checks if there are obstructions less than 3 metres from the rear of the car.

- (c) State the rules that need to be added to **Figure 1** to test this condition.

[2]

Information similar to that presented in **Figure 1** could be used to construct decisions and conditions in program design (see **Figure 2**).

Figure 2: Identifiers for car motion rules

Identifier	Description
F	Distance in metres to the vehicle in front
S	Speed of car in kilometres per hour
T	Travelling in a forward direction

- (d) Determine the value of the following expression given that the input values for F, S and T are:

F = 5

S = 30

T = true

F >= 25 AND S >= 10 AND S <= 130 AND T = true

You must show your working.

[2]

- (e) Construct an algorithm in pseudocode that repeats the following steps while the car is moving:

- Input the value for the distance from the vehicle in front.

- Input the value for the speed of the car.
- Check the inputs and notify the user if either the distance from the car in front is less than 20 metres or if the speed of the car is more than 130 kilometres per hour.

[6]

The algorithm will only terminate when the car stops moving.

14. [Maximum mark: 3]

23N.1.SL.TZ0.5

Construct a truth table for the following expression:

A OR NOT B AND C

[3]

15. [Maximum mark: 2]

23N.1.SL.TZ0.7

Identify **three** functions of the control unit (CU) in the central processing unit (CPU).

[2]

16. [Maximum mark: 1]

23N.1.HL.TZ0.2

Define the Boolean **NAND** operator.

[1]

17. [Maximum mark: 2]

23M.1.SL.TZ1.1

Outline the function of a web browser.

[2]

18. [Maximum mark: 4]

23M.1.SL.TZ1.4

(a) Identify **one** characteristic of random access memory (RAM).

[1]

(b) Explain the use of cache memory.

[3]

19. [Maximum mark: 4] 23M.1.SL.TZ1.5
Construct a truth table for the logic expression

A NAND (B NOR C) [4]

20. [Maximum mark: 2] 23M.1.HL.TZ1.3
Outline **one** reason for using Unicode to represent data in a computer system. [2]

21. [Maximum mark: 2] 23M.1.HL.TZ1.7
Distinguish between random access memory (RAM) and read-only memory (ROM). [2]

22. [Maximum mark: 2] 22N.1.SL.TZ0.7
Each pixel on a computer screen has three colour values associated with it: red, green and blue. The range for each of the three colour values is from $0_{(10)}$ to $255_{(10)}$.

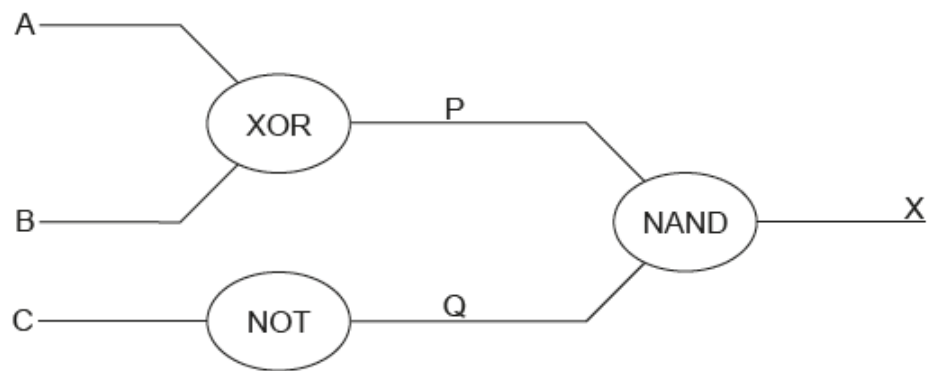
Colour values can also be represented in hexadecimal. For example, the colour blue can be represented in hexadecimal as 0000FF.

(a) State the binary representation of the colour blue. [1]

(b) State the number of colours that can be represented in each pixel on the computer screen. [1]

23. [Maximum mark: 4] 22M.1.SL.TZ0.6

Draw the truth table for the following logic circuit.



[4]